



Audio, Video, and
Iframe Embedding in
HTML

Today's Topics

- ❖ Embedding Multimedia in Web Pages
- ❖ The `<audio>` Tag
- ❖ The `<video>` Tag
- ❖ Supported Formats and Attributes
- ❖ The `<iframe>` Tag
- ❖ Embedding YouTube and Web Pages





Introduction to Multimedia Embedding

- HTML5 supports **native embedding** of media files like **audio, video,** and external content using **iframe**.
- Previously, plugins (like Flash) were required.
- These tags make multimedia:
 - Easier to implement
 - Accessible across browsers
 - Mobile-friendly
- Embedding improves interactivity, engagement, and content delivery



The <audio> Tag -

➤ Overview

➤ Used to embed sound or music files into a web page.

➤ Introduced in HTML5.

➤ Supports multiple file formats: .mp3, .ogg,

.wav

```
<audio controls>
```

```
  <source src="sound.mp3" type="audio/mpeg">
```

Your browser does not support the audio tag.

```
</audio>
```





<audio> Tag Attributes

Attribute	Description
controls	Displays play/pause UI
autoplay	Starts audio automatically on load
loop	Repeats the audio continuously
muted	Starts audio in a muted state
preload	Instructs browser how to load audio file

➤ Note: Autoplay is often blocked by modern browsers unless muted.



The <video> Tag -

Overview

➤ Allows you to embed video content directly into HTML.

➤ Eliminates the need for third-party video players.

➤ Common file formats: .mp4, .webm, .ogg

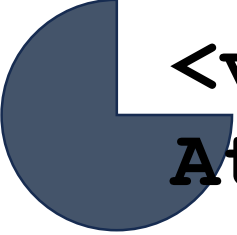
```
<video width="400" controls>
```

```
  <source src="movie.mp4" type="video/mp4">
```

```
  Your browser does not support the video tag.
```

```
</video>
```





<video> Tag Attributes

Attribute	Description
controls	Displays video controls
autoplay	Auto-plays the video
loop	Repeats video playback
muted	Mutes audio by default
poster	Specifies an image to show before play
width/height	Sets video dimensions





Multiple Source Support

- Use multiple `<source>` tags inside `<audio>` or `<video>` to provide fallbacks.

Example:

```
<video controls>  
  <source src="movie.mp4" type="video/mp4">  
  <source src="movie.webm" type="video/webm">  
  Your browser does not support this video.  
</video>
```

- The browser picks the first supported format automatically.





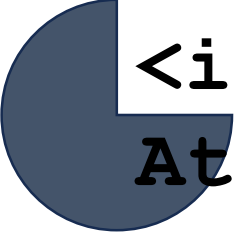
What is an <iframe>?

- An <iframe> (inline frame) embeds another HTML document inside the current page.
- Useful for:
- YouTube videos
- Maps
- Other web content (e.g., forms, documents)

Basic Syntax:

```
<iframe src="https://example.com" width="500" height="300"></iframe>
```





<iframe> Attributes

Attribute	Description
src	URL of the document to embed
width/height	Dimensions of the iframe
title	Accessibility description
frameborder	Border around the frame (deprecated)
allowfullscre en	Allows fullscreen display





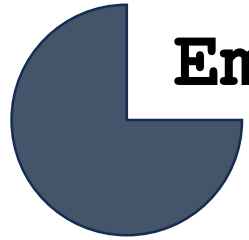
Embedding YouTube Videos with Iframe

Example:

```
<iframe  
  width="560"  
  height="315"  
  src="https://www.youtube.com/embed/VIDEO_ID"  
  title="YouTube video player"  
  frameborder="0"  
  allowfullscreen>  
</iframe>
```

- Replace VIDEO_ID with the actual YouTube video ID.
- YouTube provides an "Embed" option to auto-generate this code.



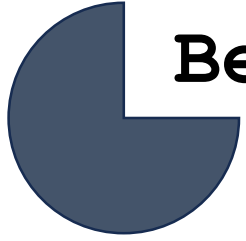


Embedding Google Maps

```
<iframe  
  src="https://www.google.com/maps/embed?pb=..."  
  width="600"  
  height="450"  
  style="border:0;"  
  allowfullscreen=""  
  loading="lazy">  
</iframe>
```

- Embeds interactive map into your website.
- Ideal for contact or location sections.





Best Practices for Media and Iframes

- Use **multiple formats** for audio/video for browser support.
- Use the controls attribute for user access.
- Don't autoplay unless muted (or user-initiated).
- Always provide **fallback content**.
- Avoid overuse of iframes—they can slow down performance.
- Use descriptive title for iframes to improve accessibility.



That's All for Today
Guys!!

Any Queries?

